AI BlogNest API – Phase Document

Project Title

AI BlogNest API

Project Overview

AI BlogNest API is a RESTful backend application designed to simplify blog management using Artificial Intelligence. The system allows users to create, edit, publish, and manage blog posts while integrating AI-powered features such as automatic blog generation and content summarization. It provides secure authentication, role-based authorization, and efficient database management through MongoDB.

Phase 1: Requirement Analysis

Objectives

- Understand the requirements of the blog management system.
- Identify the functional and non-functional requirements.
- Define user roles and permissions.
- Plan AI integration for content generation and summarization.

Functional Requirements

- User Registration and Login
- JWT Authentication
- Role-Based Authorization (Admin, Author, Reader)
- Blog CRUD Operations
- Comment Management
- AI Blog Generation

- AI Content Summarization
- Search and Filter Blogs

Non-Functional Requirements

- High Performance
- Secure Authentication
- Data Integrity
- Scalability
- Easy Maintenance
- REST API Standards

Phase 2: System Design

Architecture

The application follows a layered architecture consisting of:

- Client/API Interface (Postman or Frontend)
- Express Server
- Authentication Middleware
- Role Authorization Middleware
- Controller Layer
- Service Layer
- AI Service
- MongoDB Database

Database Collections

- Users
- Blogs
- Comments
- Categories

API Endpoints

Authentication

- POST /api/auth/register
- POST /api/auth/login

Blog

- GET /api/blogs
- GET /api/blogs/:id
- POST /api/blogs
- PUT /api/blogs/:id
- DELETE /api/blogs/:id

AI

- POST /api/ai/generate-blog
- POST /api/ai/summarize

Phase 3: Development

Technologies Used

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- bcrypt.js
- OpenAI/Gemini API (